COMP0222 Simultaneous Localisation and Mapping

Lecture 05: Graphical Models and Factor Graphs for Linear Systems

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Structure

- The Optimal (and Unimplementable) Bayesian Solution
- What Geomatics Engineers Do…
- Graphical Models and Factor Graphs
- Formulating Recursive Estimation in a Factor Graph

The Optimal Bayesian Solution

- **The Optimal (and Unimplementable) Bayesian Solution**
- ***What Geomatics Engineers Do…***
- ***Graphical Models and Factor Graphs***
- ***Formulating Recursive Estimation in a Factor Graph***

Bayesian Approach to Estimation

- So far we've only considered the Kalman filter in terms of means and covariance and linear update rules
- However, it is possible to construct the optimal solution to the filtering and estimation problem using a full characterization of the PDF and Bayes Rule
- This Bayes-optimal estimator is guaranteed to be optimal

Problem Setup

- For the moment we won't look at the SLAM context
- The process model is

$$\mathbf{x}_{k+1} = \mathbf{f}\left[\mathbf{x}_k, \mathbf{u}_{k+1}, \mathbf{v}_{k+1}\right]$$

- The observation model is

$$\mathbf{z}_{k+1} = \mathbf{h}\left[\mathbf{x}_{k+1}, \mathbf{u}_{k+1}, \mathbf{w}_{k+1}\right]$$

Bayesian Filtering Problem

- The goal is to estimate the *probability density function* of the state of interest,

$$f\left(\mathbf{x}_k \middle| \mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0\right)$$

- Once we've got this, we can derive any quantity and precisely and correctly from our system

Bayesian Filtering Problem

- To simplify our equations, we'll introduce the notation

$$\mathbb{I}_k = \{\mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0\}$$

- Therefore, we can rewrite the filtering problem as

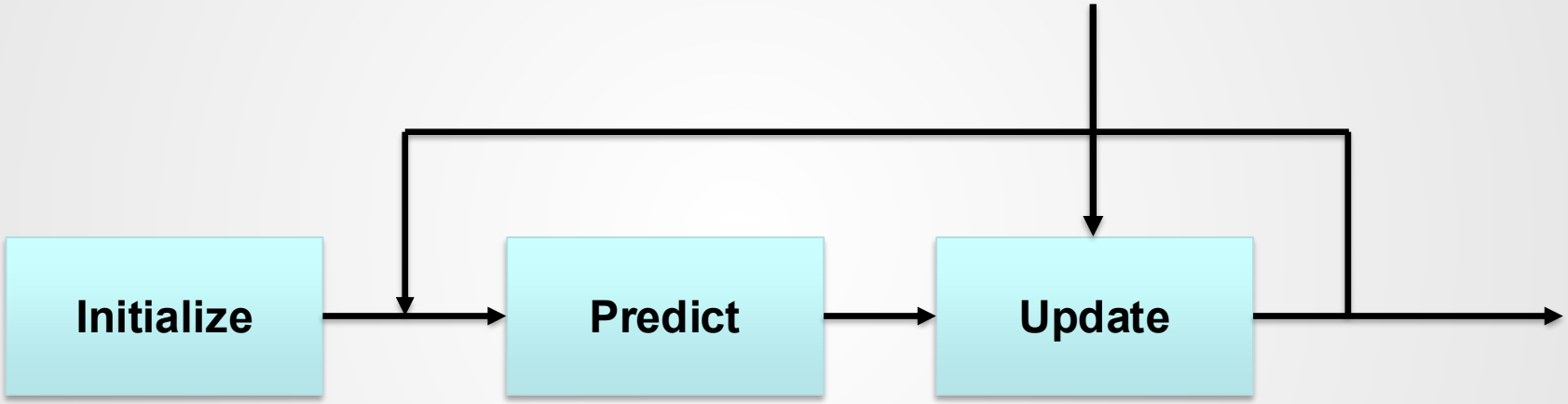
$$f\left(\mathbf{x}_k \middle| \mathbb{I}_k\right)$$

Bayesian Filtering Problem and the PDF

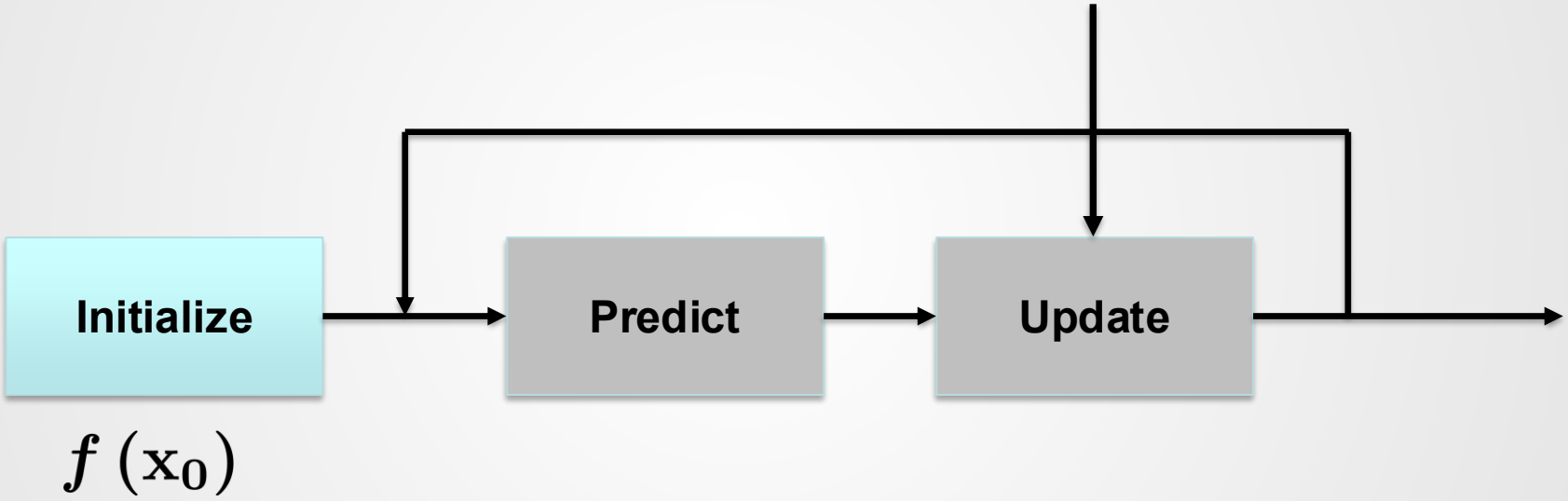
- For example, the mean is given by

$$\hat{\mathbf{x}}_{\boldsymbol{k}} = \int \mathbf{x}' f\left(\mathbf{x}' | \mathbb{I}_{\boldsymbol{k}}\right) \mathrm{d}\mathbf{x}'$$

Overview of the Bayesian Filter



Bayesian Filter Initialization



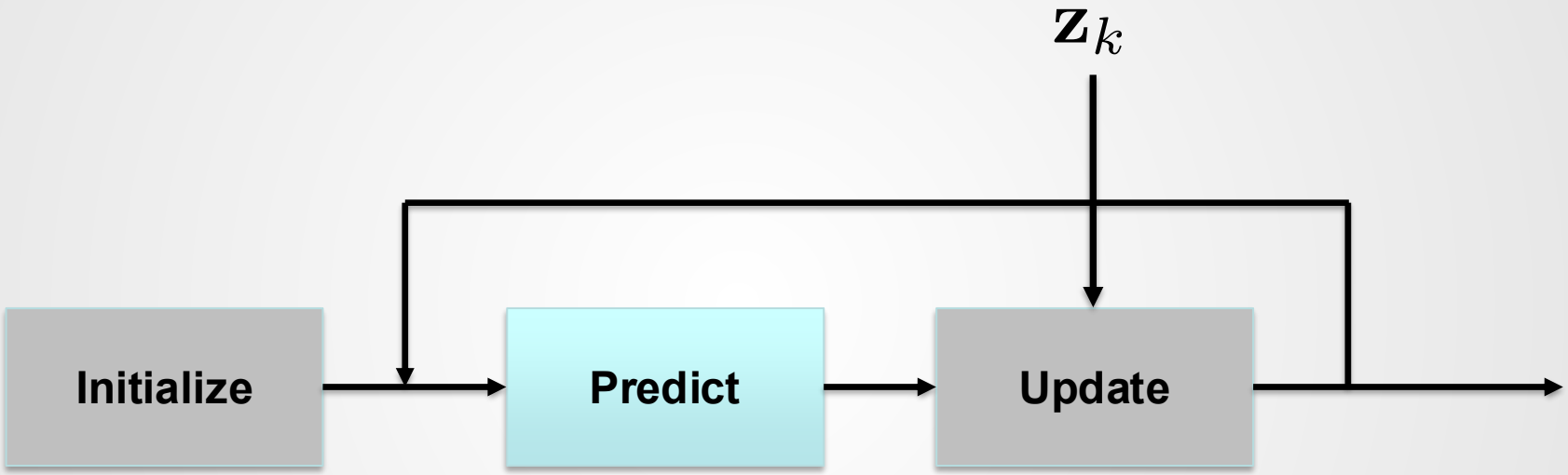
Initialization

- As with the Kalman filter, we have to start with an initial state
- For example, if the robot starts at a known location, the prior is the delta function

$$f\left(\mathbf{x}_0\right) = \delta_{\mathbf{x}_I}\left(\mathbf{x}_0\right)$$

$$\underline{x_0 = x_I}$$

Prediction Step



$$f\left(\mathbf{x}_{k+1} \middle| \mathbb{I}_k, \mathbf{u}_{k+1}\right)$$

Prediction Step

- The prediction is computed using the Chapman-Kolmogorov Equation
- However, you have to make sure that you use the *correct* one

Wrong Chapman Kolmogorov Equation

Right Chapman Kolmogorov Equation

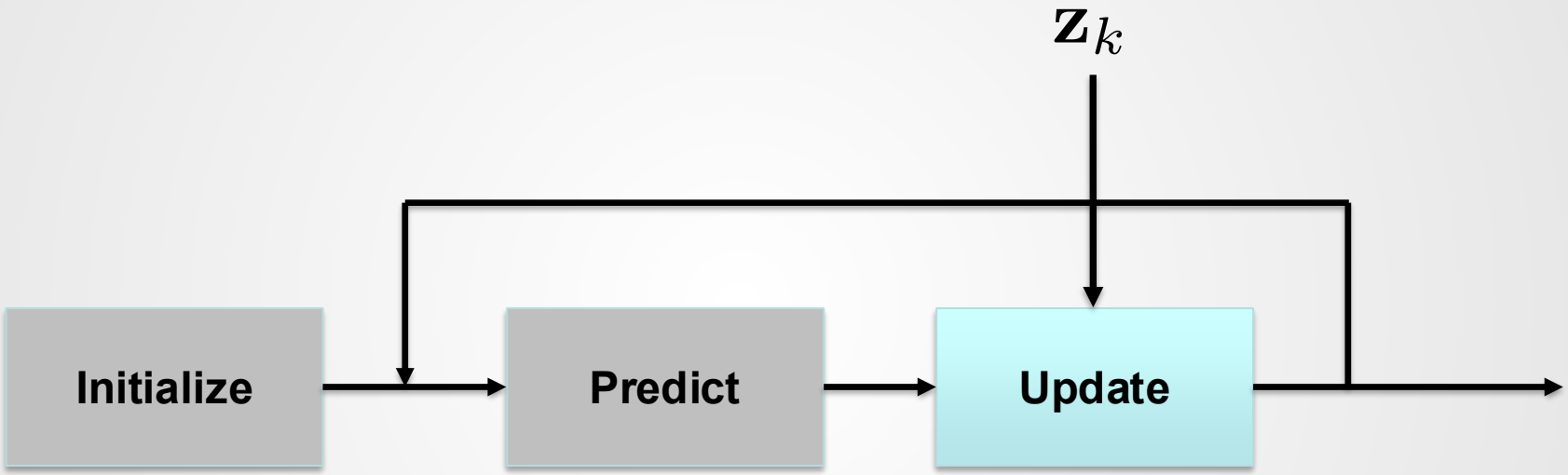
$$f\left(\mathbf{x}_{k+1}|\mathbb{I}_k, \mathbf{u}_{k+1}\right) = \int f\left(\mathbf{x}_{k+1}|\mathbf{x}', \mathbf{u}_{k+1}\right) f\left(\mathbf{x}'|\mathbb{I}_k\right) \mathrm{d}\mathbf{x}'$$

The Chapman-Kolmogorov Equation

- The equation produces the *exact solution*
- It can fully and completely propagate the effects such as multi-modality and effects of angles
- Unfortunately, it's impossible to compute in almost all cases
- The only one you can work out is for linear Gaussian systems, where the distribution is Gaussian and whose mean and covariance is the same as that computed by the KF

Measurement Update

REDRAW



$$f\left(\mathbf{x}_{k+1} \middle| \mathbb{I}_{k+1}\right)$$

Measurement Update

- **We use Baye's Rule,**

$$f\left(\mathbf{x}_{k+1}|\mathbb{I}_{k+1}\right) = \frac{f\left(\mathbf{z}_{k+1}|\mathbf{x}_{k+1}\right) f\left(\mathbf{x}_{k+1}|\mathbb{I}_{k}, \mathbf{u}_{k+1}\right)}{f\left(\mathbf{z}_{k+1}|\mathbb{I}_{k}, \mathbf{u}_{k+1}\right)}$$

Normalization Constant

- **The normalization constant (evidence) is given by**

$$f\left(\mathbf{z}_{k+1} \middle| \mathbb{I}_k, \mathbf{u}_{k+1}\right) = \int f\left(\mathbf{z}_{k+1} \middle| \mathbf{x}'\right) f\left(\mathbf{x}' \middle| \mathbb{I}_k, \mathbf{u}_{k+1}\right) \mathrm{d}\mathbf{x}'$$

Computing the Normalization Constant

- Surprisingly enough…
  - You can't
- About the only case you can compute it is for a linear Gaussian filter
- In that case, you can show that the posterior is a Gaussian distribution whose mean and covariance are the same as those computed by the KF update equations

So Where Next?

- If the Kalman filter doesn't work and the optimal solution can't be implemented what can we do???
- Fortunately, it turns out that people have been building maps for a very long time
- These provide a way to reformulate the problem in a form that *works*

What Geomatics Engineers Do…

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UNIVERSITY COLLEGE LONDON
Department of Geomatic Engineering

Evaluation of the ADS40 and ALS50 for automated building reconstruction

Nikolaos Kokkas

M.Sc. in Photogrammetry with Remote Sensing
September 2005

Project supervisors:

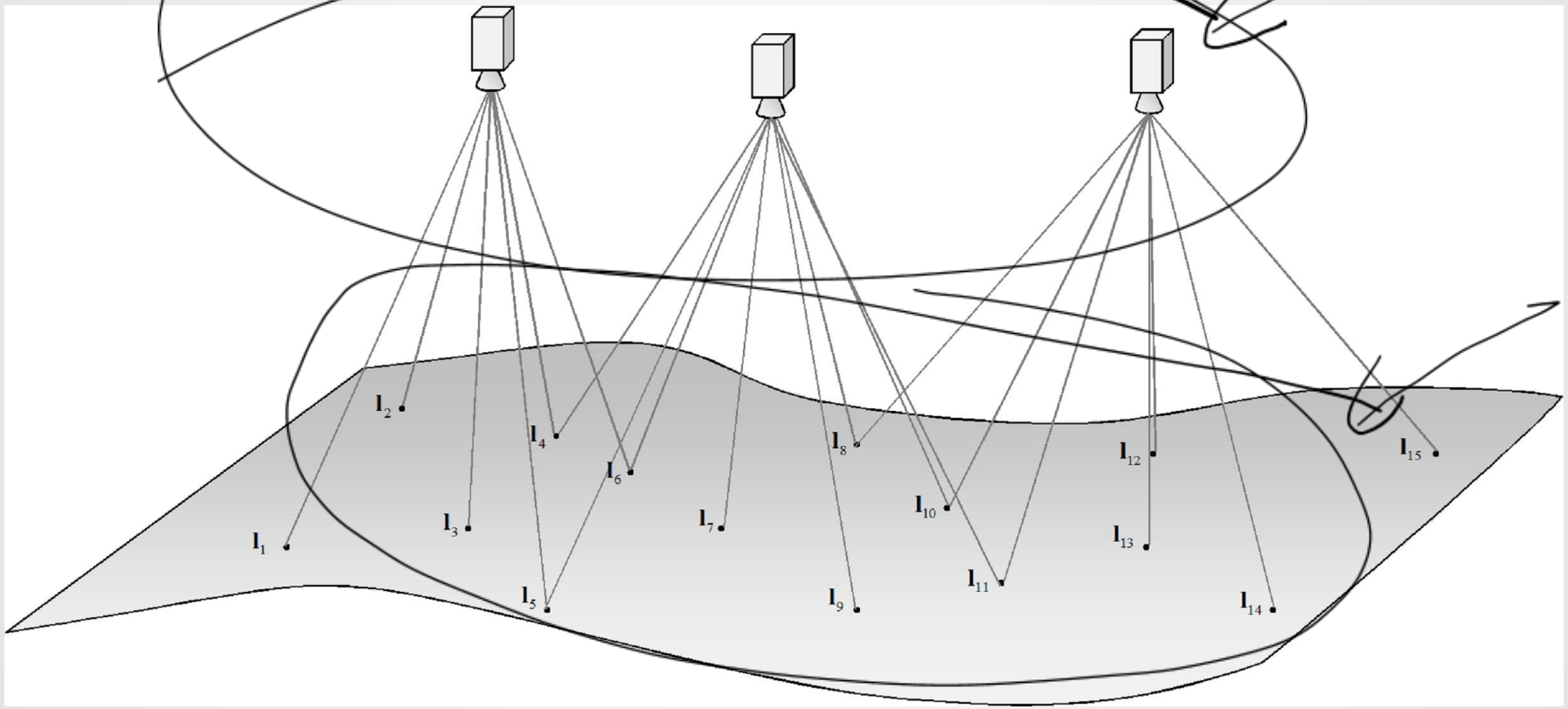
Professor I. J. Dowman

Dr. Stuart Robson

Aerial Photography and Map Making

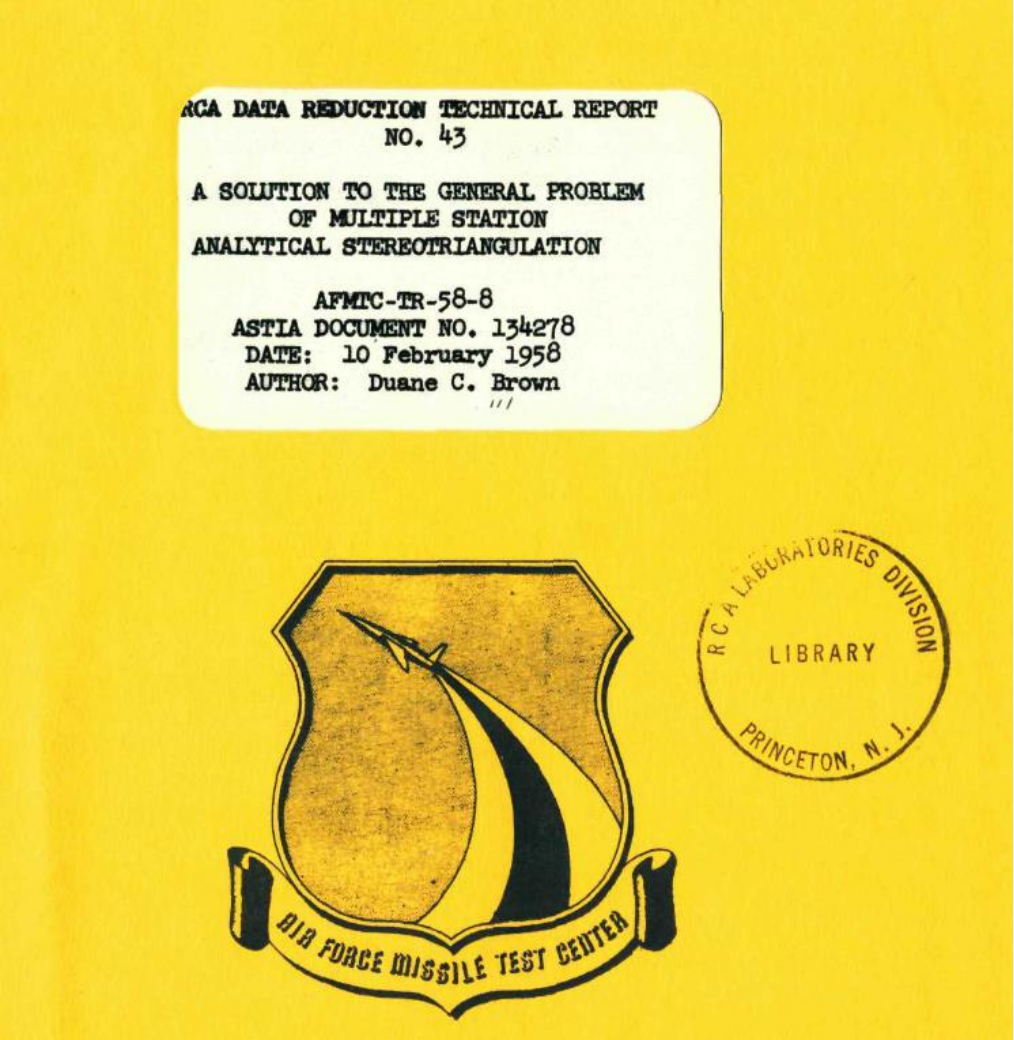
Part of a photographic plate used for map making from the 1950s

Using Photogrammetry for Map Making



From http://www.geodetic.com/v-stars/what-is-photogrammetry.aspx

Using Photogrammetry for Map Making



RCA DATA REDUCTION TECHNICAL REPORT
NO. 43

A SOLUTION TO THE GENERAL PROBLEM
OF MULTIPLE STATION
ANALYTICAL STEREOTRIANGULATION

AFMTC-TR-58-8
ASTIA DOCUMENT NO. 134278
DATE: 10 February 1958
AUTHOR: Duane C. Brown

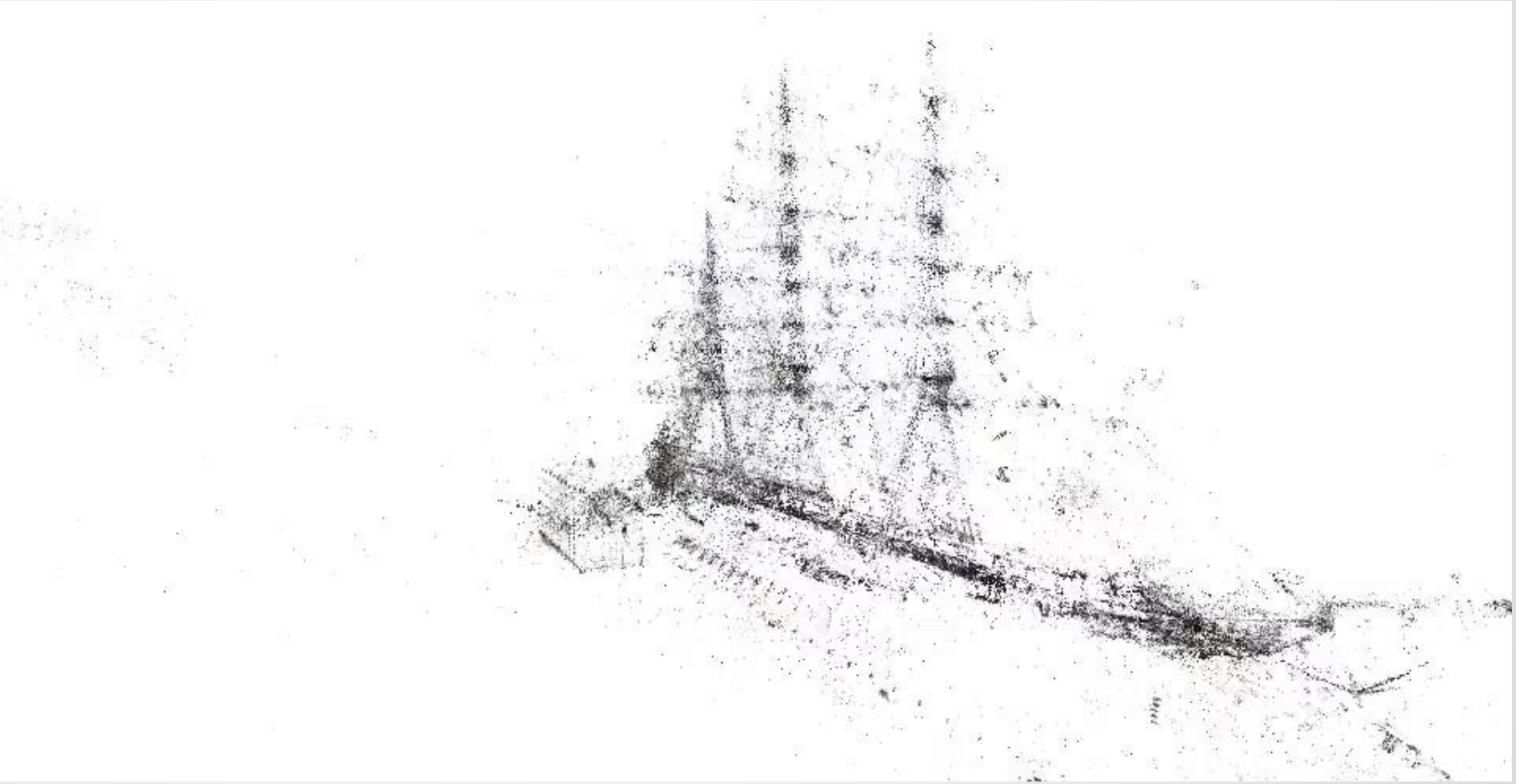
RCA LABORATORIES DIVISION
LIBRARY
PRINCETON, N. J.

AIR FORCE MISSILE TEST CENTER

Structure from Motion

Bundler: Structure from Motion (SfM) for Unordered Image Collections (2006)

COLMAP Building a Map of the Cutty Sark



How Are These Maps Built?

- These are *Structure from Motion systems* which use *Bundle Adjustment*
- Bundle adjustment thinks about mapping and estimation in a very different way:
  - The way the problem is formulated is different
  - The way we compute the estimate from the problem is different

The Original SLAM Equation

$$f\left(\mathbf{x}_k, \mathbf{m} \middle| \mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0\right)$$

Current pose of the platform

Map

$$\mathbf{m} = \left\{\mathbf{m}^1, \cdots, \mathbf{m}^{N_k}\right\}$$

Set of all observations

$$\mathbf{Z}_{0:k} = \{\mathbf{Z}_0, \ldots, \mathbf{Z}_k\}$$

Set of all control inputs

$$\mathbf{U}_{0:k} = \{\mathbf{u}_0, \ldots, \mathbf{u}_k\}$$

Initial conditions

The Bundle Adjustment Equivalent

$$f\left(\mathbf{x}_{1:k}, \mathbf{m} \middle| \mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0\right)$$

History **of the pose of the platform**

Map

$$\mathbf{m} = \left\{\mathbf{m}^1, \cdots, \mathbf{m}^{N_k}\right\}$$

Set of all observations

$$\mathbf{Z}_{0:k} = \{\mathbf{Z}_0, \ldots, \mathbf{Z}_k\}$$

Set of all control inputs

$$\mathbf{U}_{0:k} = \{\mathbf{u}_0, \ldots, \mathbf{u}_k\}$$

Initial conditions

The Original SLAM Equation

$$\hat{\mathbf{x}}_{\boldsymbol{k}|\boldsymbol{k}} = \mathbb{E}\left[\mathbf{x}_{\boldsymbol{k}} | \mathbb{I}_k\right]$$

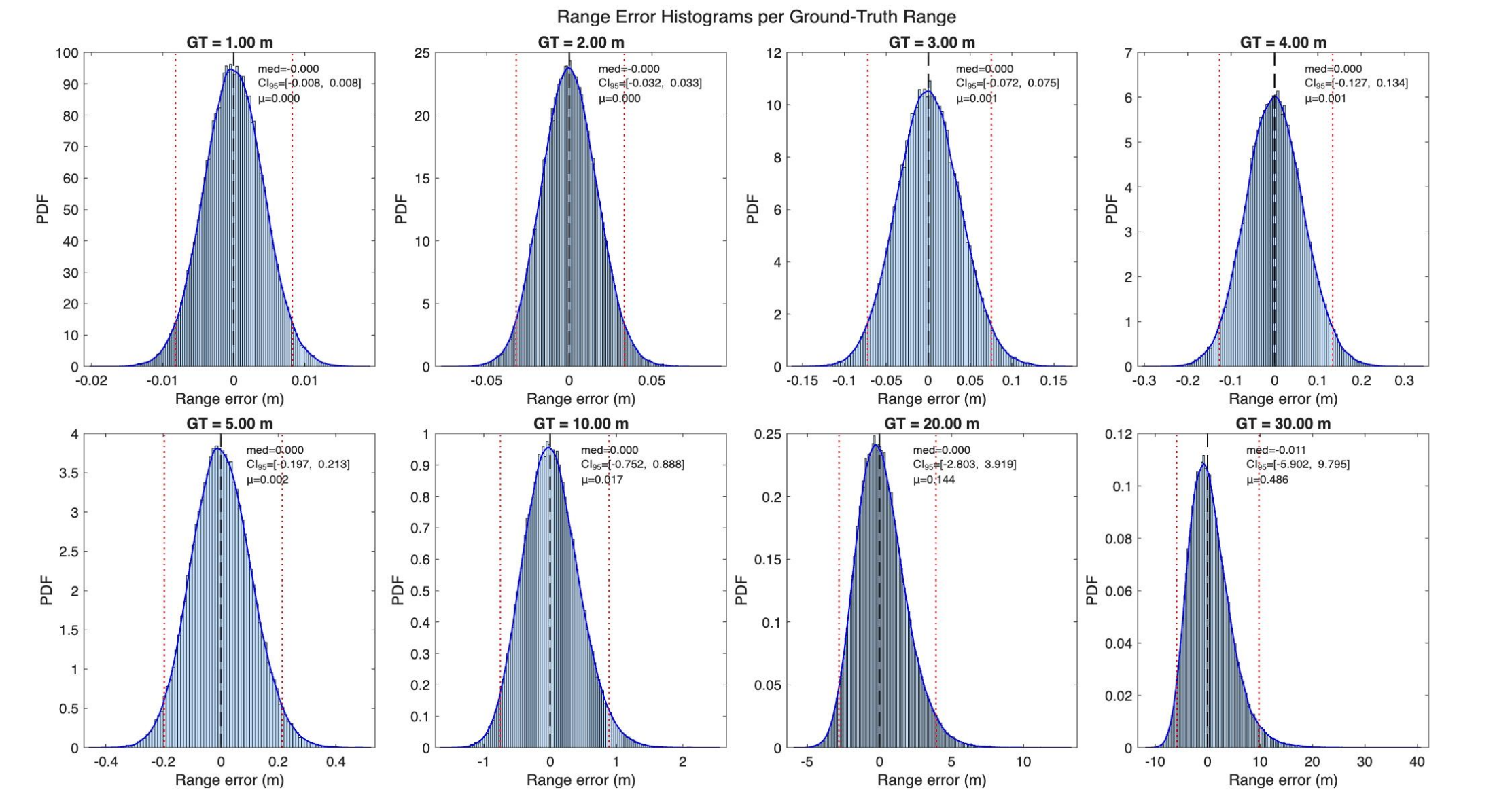
$$\mathbf{P}_{\boldsymbol{k}|\boldsymbol{k}} = \mathbb{E}\left[\tilde{\mathbf{x}}_{\boldsymbol{k}|\boldsymbol{k}} \tilde{\mathbf{x}}_{\boldsymbol{k}|\boldsymbol{k}}^{\top} | \mathbb{I}_k\right]$$

The Bundle Adjustment Equivalent

$$\hat{\mathbf{x}}_{1:k} = \arg\max_{\mathbf{x}_{1:k}} f\left(\mathbf{x}_{1:k} | \mathbb{I}_k\right)$$

Maximum A Posteriori Estimation

Maximum A Posteriori Estimation vs. the Mean


Range Error Histograms per Ground-Truth Range
GT = 1.00 m
med=-0.000
CI95=[-0.008, 0.008]
μ=0.000
GT = 2.00 m
med=-0.000
CI95=[-0.032, 0.033]
μ=0.000
GT = 3.00 m
med=0.000
CI95=[-0.072, 0.075]
μ=0.001
GT = 4.00 m
med=0.000
CI95=[-0.127, 0.134]
μ=0.001
GT = 5.00 m
med=0.000
CI95=[-0.197, 0.213]
μ=0.002
GT = 10.00 m
med=0.000
CI95=[-0.752, 0.888]
μ=0.017
GT = 20.00 m
med=0.000
CI95=[-2.803, 3.919]
μ=0.144
GT = 30.00 m
med=-0.011
CI95=[-5.902, 9.795]
μ=0.486
PDF
Range error (m)

So How Do we Implement This?

- We need to come up with the probabilistic machinery which lets us describe and solve these problems
- It turns out the nature of the systems we are dealing with can be described using Markov Chains
- Before we do this, though, we are going to introduce graphical models and factor graphs
- These are a very convenient way to picture the system we are dealing with

Graphical Models and Factor Graphs

- *The Optimal (and Unimplementable) Bayesian Solution)*
- *What Geomatics Engineers Do…*
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Graphical Models

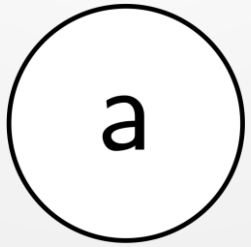
- Graphical models are a way of decomposing large, horrible probabilistic equations into smaller, nicer ones
- They use conditional probability and conditional independence
- They make it possible to have highly “componentised” equations which are very general, flexible and extensible

Graphical Models

- **The graph uses the following:**
  - **The *vertices* (or *nodes* or *variables*) denote the events whose probabilities we want to calculate**
  - **The *edges* specify conditional probabilities linking those events**

Specifying a Latent Variable

- We first consider how to specify a single variable, a
- Latent means we don't know the value of the variable, and we want to estimate it
- The graph consist of just the vertex corresponding to that variable

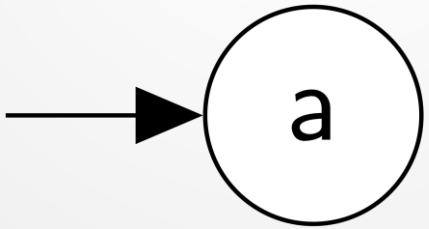


Specifying the Distribution on that Variable

- We'd now like to encode the probability distribution,

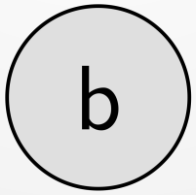
$$f\,(\mathrm{a})$$

- We do this by adding an edge to the variable node,



Graphical Representation of Observed Variables

- Graphical models let you introduce *observed variables*
- Their values are known (observed), so they aren't random
- This is how we feed stuff like control inputs and observations into the system
- These are represented by the greyed out nodes



Reminder on Independence

- Consider the joint probability (density) over events a and b,

$$f\left(\mathrm{a}, \mathrm{b}\right)$$

- If the events are *independent* of one another,

$$f\left(\mathrm{a}, \mathrm{b}\right) = f\left(\mathrm{a}\right) f\left(\mathrm{b}\right)$$

Reminder on Conditional Probability

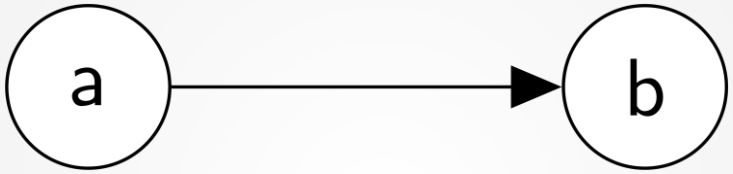
- In general, events are not independent
- We often model this dependency using the *conditional probability*
- This is defined such that

$$f(\mathbf{a}, \mathbf{b}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})$$

Reminder on Conditional Probability

$$f(\mathbf{a}, \mathbf{b}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})$$

Graphical Representation



$$f(\mathbf{a}, \mathbf{b}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})$$

Reminder on the Probabilistic Chain Rule

- Now consider the joint probability over *three* events,

$$f\left(\mathbf{a}, \mathbf{b}, \mathbf{c}\right)$$

- If these events were independent,

$$f\left(\mathbf{a}, \mathbf{b}, \mathbf{c}\right) = f\left(\mathbf{a}\right) f\left(\mathbf{b}\right) f\left(\mathbf{c}\right)$$

Reminder on the Probabilistic Chain Rule

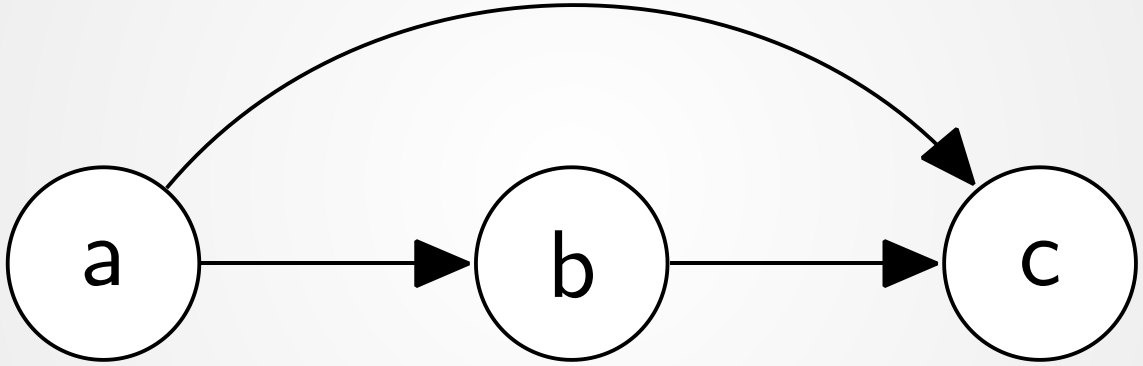
- **If the events are not *independent*, we can recursively build them up using conditional probabilities that involve more terms**

$$f(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})\, f(\mathbf{c}|\mathbf{a}, \mathbf{b})$$

Reminder on the Probabilistic Chain Rule

$$f(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})\, f(\mathbf{c}|\mathbf{a}, \mathbf{b})$$

Graphical Model Representation of the 3 Variable Case



$$f(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})\, f(\mathbf{c}|\mathbf{a}, \mathbf{b})$$

Reminder on Conditional Independence

- We can simplify this if we have *conditional independence*
- Suppose events b and c are not independent of one another
- Therefore,

$$f(\mathbf{b}, \mathbf{c}) \neq f(\mathbf{b})\, f(\mathbf{c})$$

Reminder on Conditional Independence

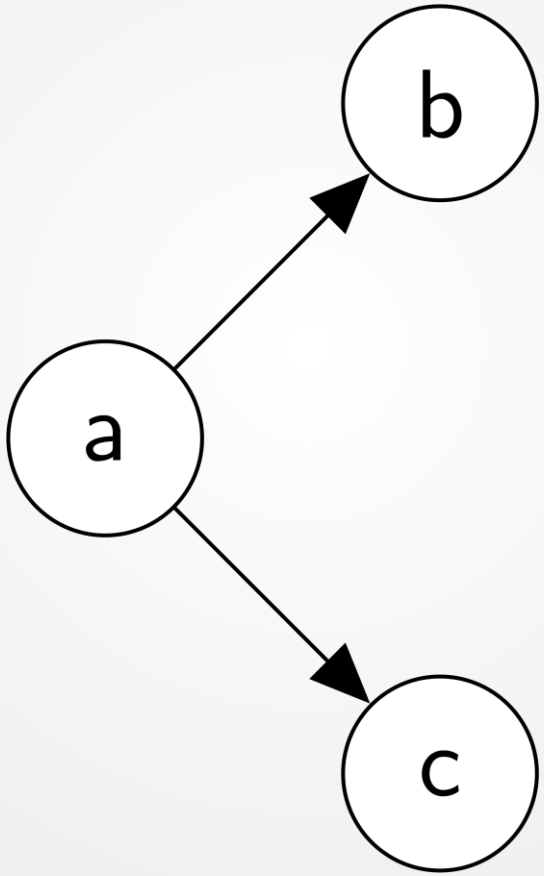
- However, b and c are conditionally independent of one another if, conditioning on a, it's true that

$$f(\mathbf{b}, \mathbf{c}|\mathbf{a}) = f(\mathbf{b}|\mathbf{a})\, f(\mathbf{c}|\mathbf{a})$$

Reminder on Conditional Independence

$$f\left(\mathbf{b}, \mathbf{c} | \mathbf{a}\right) = f\left(\mathbf{b} | \mathbf{a}\right) f\left(\mathbf{c} | \mathbf{a}\right)$$

Graphical Model Representation for Conditional Independence



Joint and Conditional Probabilities for 5 Variable Case

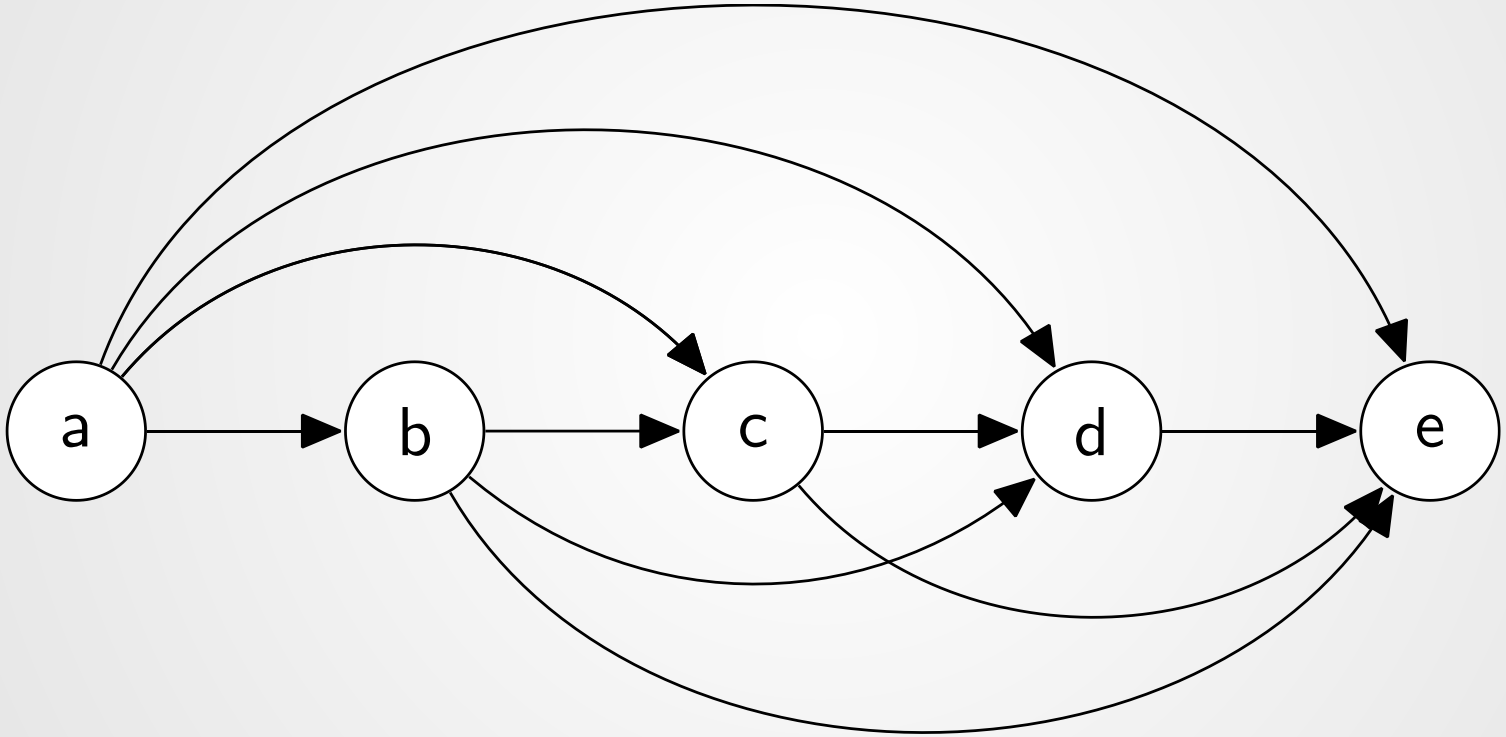
- Now let's consider *five* random variables

$$f\left(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}\right)$$

- Using conditional probabilities, this is

$$f\left(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}\right) = f\left(\mathbf{a}\right) f\left(\mathbf{b}|\mathbf{a}\right) f\left(\mathbf{c}|\mathbf{a}, \mathbf{b}\right) \times f\left(\mathbf{d}|\mathbf{a}, \mathbf{b}, \mathbf{c}\right) f\left(\mathbf{e}|\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}\right)$$

Graphical Model Representation in 5 Variable Case



$$f(\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d},\mathbf{e}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})\, f(\mathbf{c}|\mathbf{a},\mathbf{b})\, f(\mathbf{d}|\mathbf{a},\mathbf{b},\mathbf{c})\, f(\mathbf{e}|\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d})$$

Graphical Model and Independence

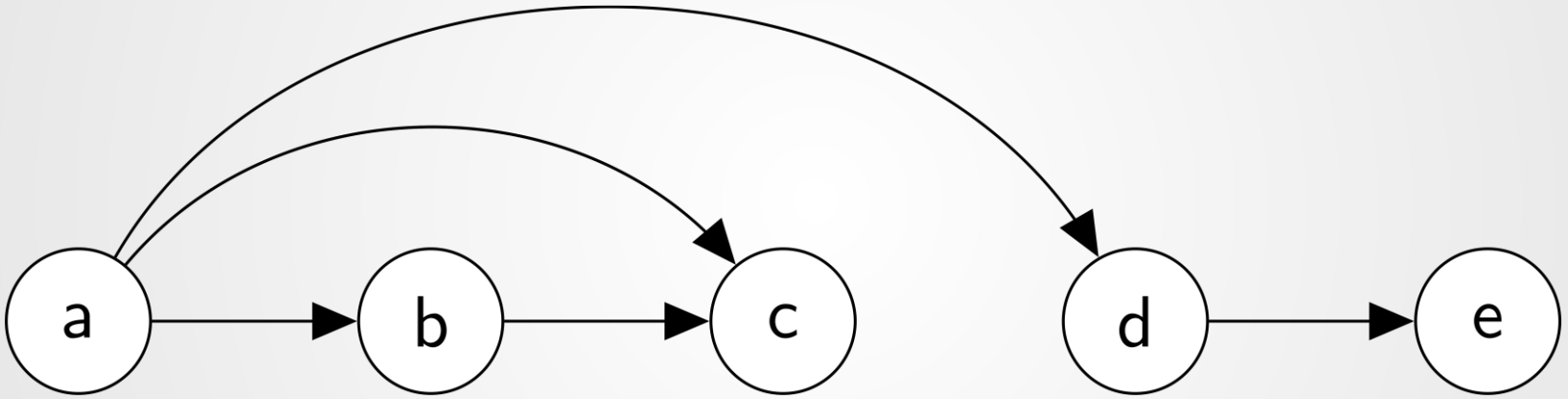
- This structure can be used to describe any joint probability of random variables
- However, why is it useful?
- The reasons are:
  - We can often model the system structure more easily
  - Doing this, we can exploit the *sparsity* between random variables

Sparsity and Conditional Probabilities

- Suppose that d is conditionally independent of b, and c and e is independent of a,b,c
- In this case

$$f(\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d},\mathbf{e}) = f(\mathbf{a})\, f(\mathbf{b}|\mathbf{a})\, f(\mathbf{c}|\mathbf{a},\mathbf{b}) \times f(\mathbf{d}|\mathbf{a})\, f(\mathbf{e}|\mathbf{d})$$

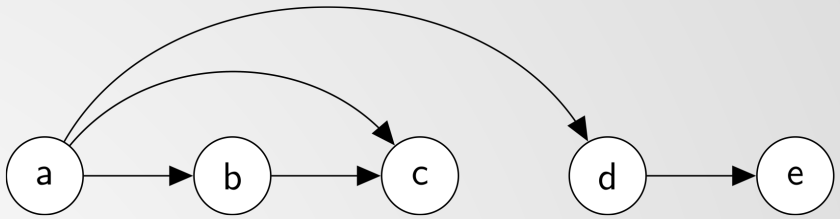
Graphical Model and Independence



Factor Graphs

- A factor graph is a general way to write down one function as a product of other functions
- It has two types of nodes:
  - *Factor nodes* represent functions which get multiplied together
  - *Variable nodes* are the values we want to estimate
- Variable nodes only connect to factor nodes, and factor nodes only connect to variable nodes

Factor Graphs



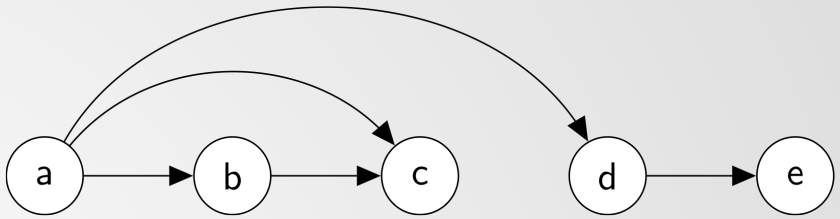
- **Consider the five variable case**

$$\mathbf{X} = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}\}$$

- **We can write the joint probability as the product of functions over *different* subsets of these variables**

$$g(\mathbf{X}) = \prod_{j=1}^{m} g_j(\mathbf{S}_j)$$

Factor Graphs



$$\mathbf{S}_1 = \{\mathbf{a}\} \qquad g_1(\mathbf{S}_1) = f(\mathbf{a})$$

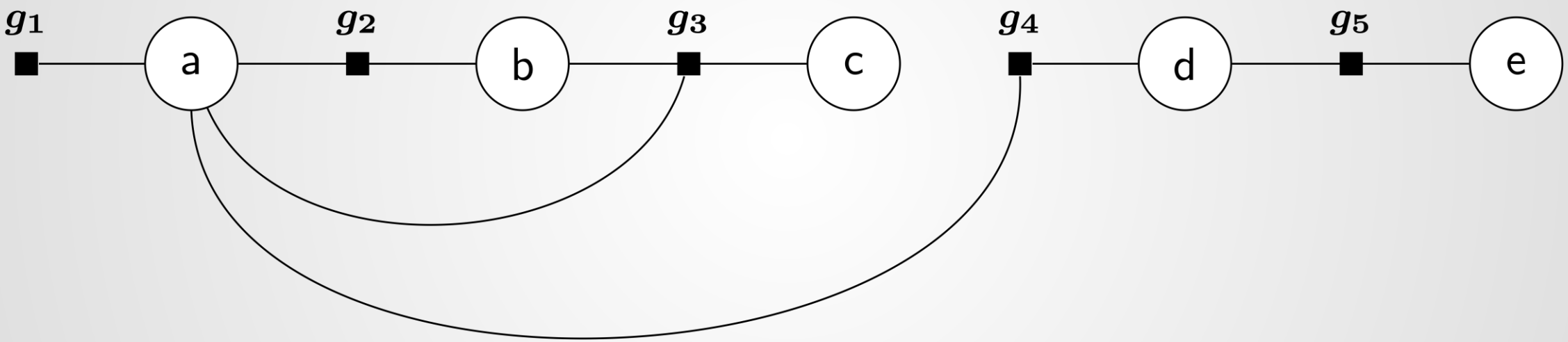
$$\mathbf{S}_2 = \{\mathbf{a}, \mathbf{b}\} \qquad g_2(\mathbf{S}_2) = f(\mathbf{b}|\mathbf{a})$$

$$\mathbf{S}_3 = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \qquad g_3(\mathbf{S}_3) = f(\mathbf{c}|\mathbf{a}, \mathbf{b})$$

$$\mathbf{S}_4 = \{\mathbf{a}, \mathbf{d}\} \qquad g_4(\mathbf{S}_4) = f(\mathbf{d}|\mathbf{a})$$

$$\mathbf{S}_5 = \{\mathbf{d}, \mathbf{e}\} \qquad g_5(\mathbf{S}_5) = f(\mathbf{e}|\mathbf{d})$$

Factor Graphs



Summarizing Graphical Models and Factor Graphs

- Independence lets us break down a joint probability into lots of conditional probabilities
- These can be visualized in graphical models and factor graphs
- Let's now see how we can apply these to the problems we are interested in
- We'll start with a tracking problem, and move onto SLAM next week

Formulating a Tracking Problem using a Factor Graph

- *The Optimal (and Unimplementable) Bayesian Solution)*
- *What Geomatics Engineers Do…*
- *Graphical Models and Factor Graphs*
- **Formulating a Tracking Problem using a Factor Graph**

Tracking Problem

- **We're going to consider the problem of writing down the conditional probability of the**

$$f\left(\mathbf{x}_{1:k} \middle| \mathbb{I}_k\right)$$

$$\mathbb{I}_k = \{\mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0\}$$

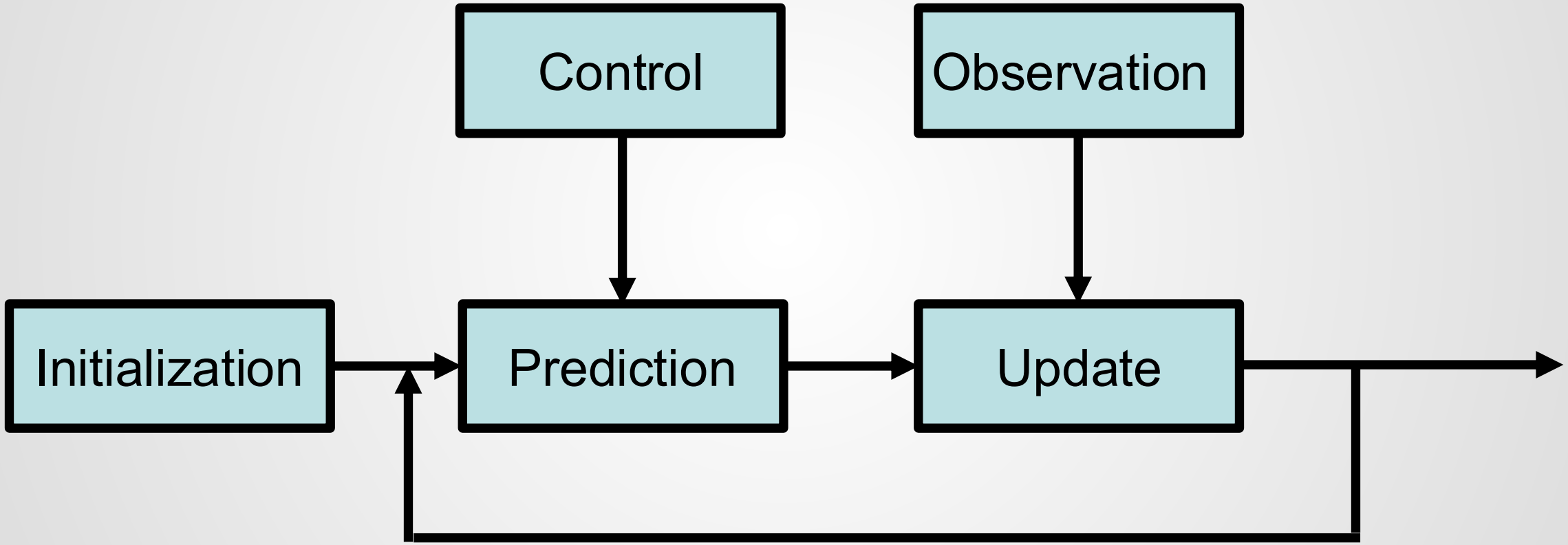
System Models

- The system equations have the familiar form

$$\mathbf{x}_{k+1} = \mathbf{F}_{k+1}\mathbf{x}_k + \mathbf{B}_{k+1}\mathbf{u}_{k+1} + \mathbf{v}_{k+1}$$
$$\mathbf{z}_{k+1} = \mathbf{H}_{k+1}\mathbf{x}_{k+1} + \mathbf{w}_{k+1}$$

- We've seen how to estimate this using a Kalman filter

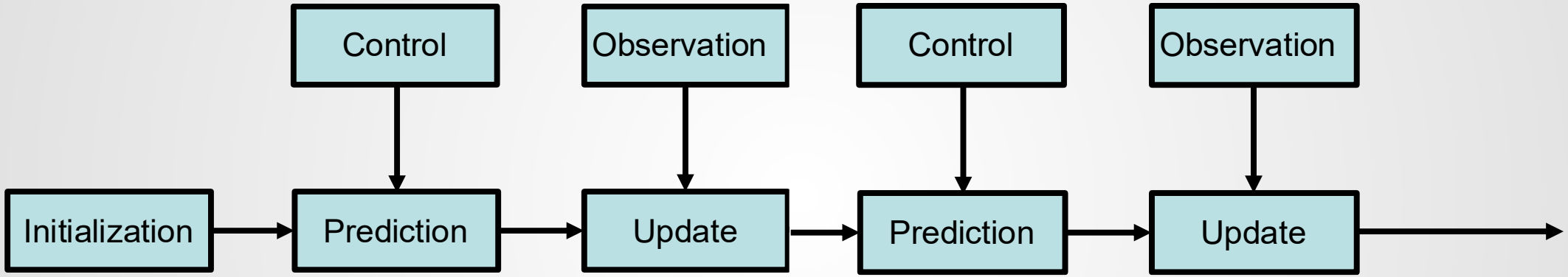
Recursive Estimator Structure for Kalman Filter



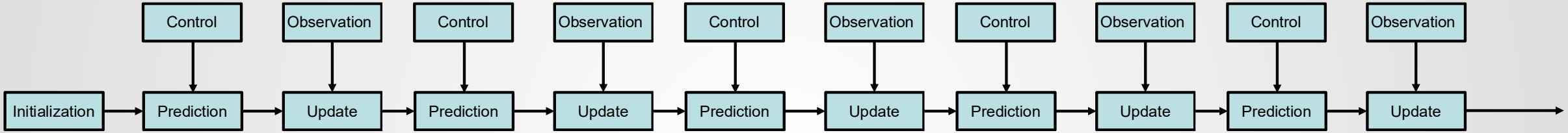
Single Time Step Kalman Filter

- The Kalman filter only stores data for *a single time*
- It does this by:
  - Taking the previous estimate
  - Predicting to the current time and throwing away the previous estimate
  - Updating the prediction and throwing away the prediction
- However, since we want to maintain the entire history, we do don't throw any estimates away
- Instead, we effectively slap all of the predict update cycles together

Structure of Multi-Time Step Estimation



Structure of Multi-Time Step Estimation



Temporal Estimator Structure

- We'll start by looking at initialization
- Then we'll look at prediction
- Finally we'll look at the update

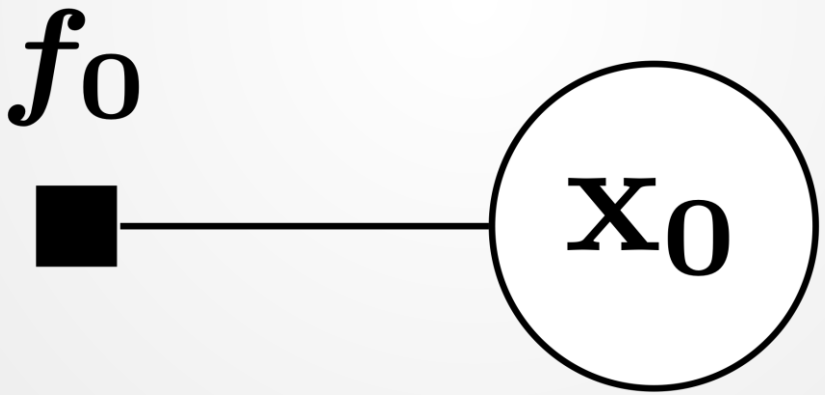
Initialization

- We need to do two things:
  - We need to specify that there is a node which is the state at time 0
  - We need to specify the distribution associate with that node

$$f\left(\mathbf{x}_0\right)$$

Graph Structure at Time Step 0

- **This produces a graph which contains a *single* vertex and a *single unitary* edge,**



Time Step 1

- **We want to estimate the joint distribution over the first *two* states given the first control input,**

$$f\left(\mathbf{x}_0, \mathbf{x}_1; \mathbf{u}_1\right)$$

Time Step 1

- **Using conditional probabilities, we can write this as**

$$f\left(\mathbf{x}_0, \mathbf{x}_1; \mathbf{u}_1\right) = f\left(\mathbf{x}_0; \mathbf{u}_1\right) f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_1\right)$$

Timestep 1

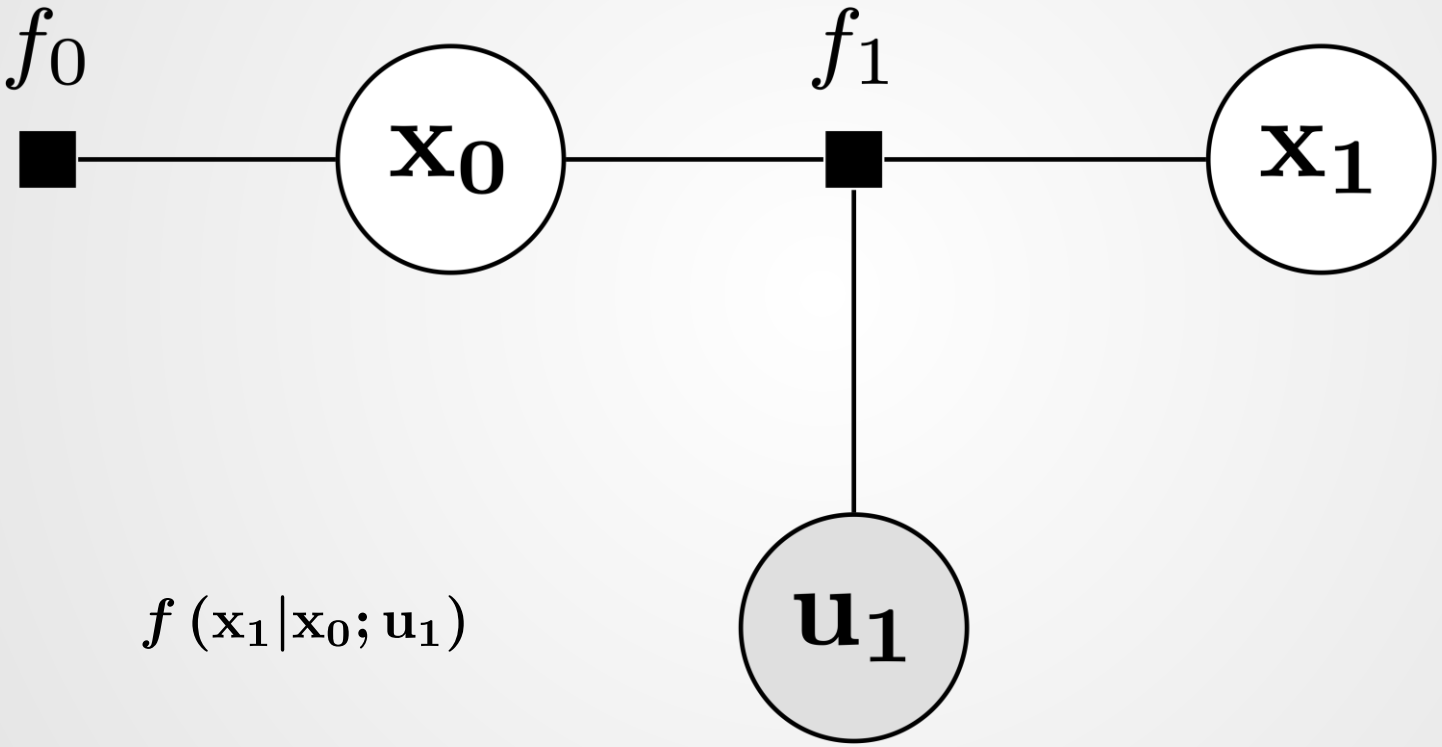
- We can make our first simplification
- Since the initial condition is *not* a function of the control input,

$$f\left(\mathbf{x}_0; \mathbf{u}_1\right) = f\left(\mathbf{x}_0\right)$$

- Therefore, the joint PDF has the form

$$f\left(\mathbf{x}_0, \mathbf{x}_1; \mathbf{u}_1\right) = f\left(\mathbf{x}_0\right) f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_1\right)$$

Factor Graph at Time Step 1



$$f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_1\right)$$

State Transition Probability

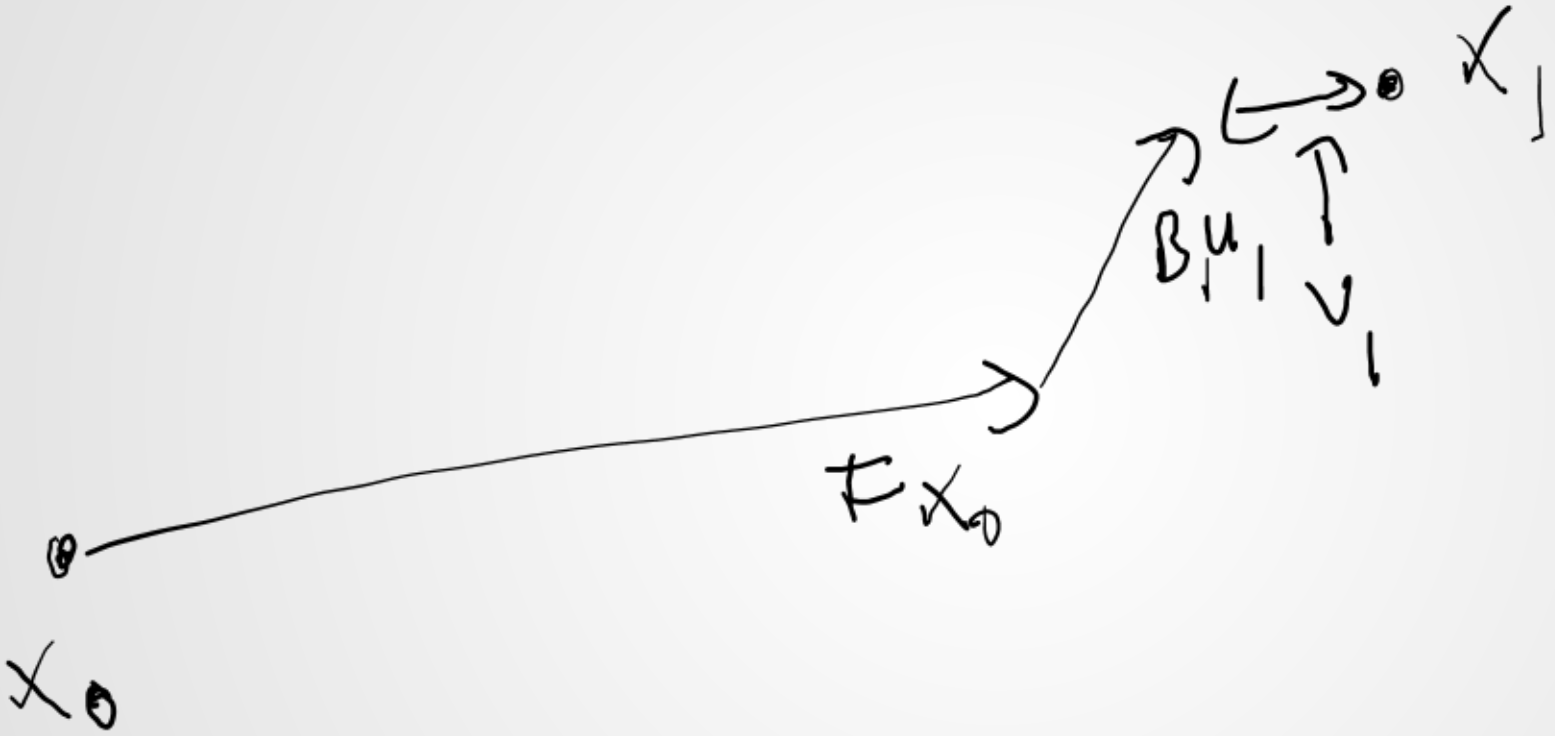
- The term

$$f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_1\right)$$

is the *state transition probability*

- It encodes the the process model

Interpreting the State Transition Probability



$$f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_1\right) \qquad \mathbf{x}_1 = \underline{\mathbf{F}_1 \mathbf{x}_0} + \underline{\mathbf{B}_1 \mathbf{u}_1} + \mathbf{v}_1$$

Value of the State Transition Probability

- We can rearrange the process model to make $\mathbf{v}_1$ the subject of the equation,

$$\mathbf{v}_1 = \underline{\mathbf{x}_1 - \mathbf{F}_1\mathbf{x}_0 - \mathbf{B}_1\mathbf{u}_1}$$

Value of the State Transition Probability

- **The state transition probability is then probability that the process noise takes the value**

$$f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_1\right) = f\left(\mathbf{v}_1 = \mathbf{x}_1 - \mathbf{F}_1 \mathbf{x}_0 - \mathbf{B}_1 \mathbf{u}_1\right)$$

Time Step 2

- **Now we want to estimate the joint distribution over the first *three* states with *two* control inputs,**

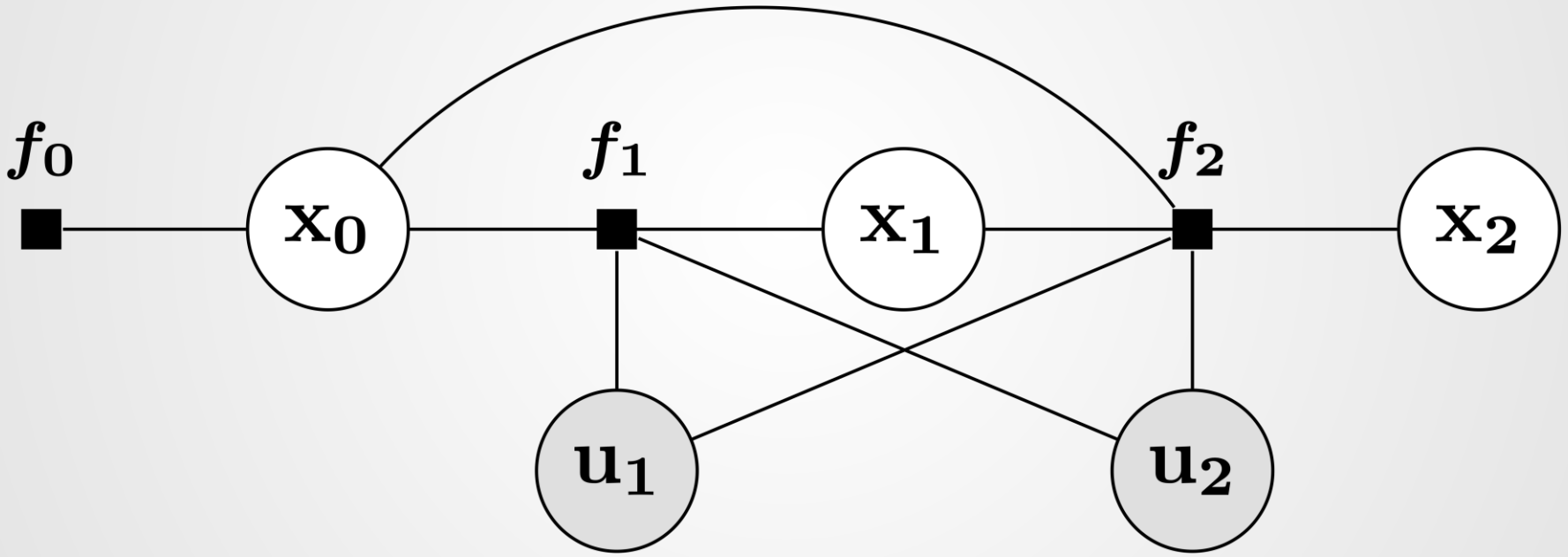
$$f\left(\underline{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2}; \mathbf{u}_{1:2}\right)$$

Time Step 2

- **The probabilistic chain rule gives us the decomposition**

$$f\left(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2\right) = f\left(\mathbf{x}_0\right) f\left(\mathbf{x}_1 | \mathbf{x}_0; \mathbf{u}_{1:2}\right) \times f\left(\mathbf{x}_2 | \mathbf{x}_1, \mathbf{x}_0; \mathbf{u}_{1:2}\right)$$

Naïve Factor Graph at Time Step 2



Form of the Process Model

- For our linear system,

$$\mathbf{x}_1 = \mathbf{F}_1\mathbf{x}_0 + \mathbf{B}_1\mathbf{u}_1 + \mathbf{v}_1$$

- Therefore, the state at time 1 *is* a function of $\mathbf{x}_0$, $\mathbf{u}_1$ and $\mathbf{v}_1$
- It is *not* a function of $\mathbf{x}_2$, $\mathbf{u}_2$ or $\mathbf{v}_2$

Form of the Process Model

- **Therefore, we can simplify the equation to**

$$f\left(\mathbf{x}_{0:2};\mathbf{u}_{1:2}\right) = f\left(\mathbf{x}_0\right)\underline{f\left(\mathbf{x}_1|\mathbf{x}_0;\mathbf{u}_1\right)} \times f\left(\mathbf{x}_2|\mathbf{x}_{0:1};\mathbf{u}_{1:2}\right)$$

Form of the Process Model

- The state at time 2 is given by

$$\mathbf{x}_2 = \mathbf{F}_2\mathbf{x}_1 + \mathbf{B}_2\mathbf{u}_2 + \mathbf{v}_2$$

- The state at time 2 *is* a function of $\mathbf{x}_1$, $\mathbf{u}_2$ and $\mathbf{v}_2$
- It is *not* a function of $\mathbf{x}_0$, $\mathbf{u}_1$ or $\mathbf{v}_1$

Simplifying the State Transition Model

- **Therefore,**

$$f\left(\mathbf{x}_2 | \mathbf{x}_{0:1}; \mathbf{u}_{1:2}\right) = f\left(\mathbf{x}_2 | \mathbf{x}_1; \mathbf{u}_2\right)$$

- **The conditional probability is given by,**

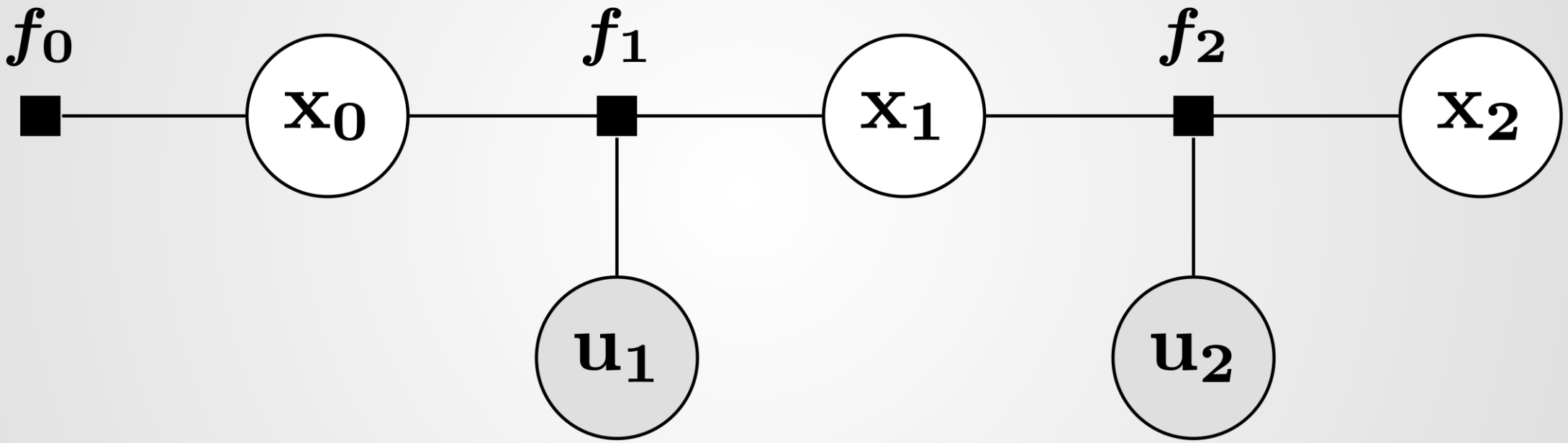
$$f\left(\mathbf{x}_2 | \mathbf{x}_1; \mathbf{u}_2\right) = f_{\mathbf{v}}\left(\mathbf{v}_2 = \mathbf{x}_2 - \mathbf{F}_2\mathbf{x}_1 - \mathbf{B}_2\mathbf{u}_2\right)$$

Form of the Process Model

- **Putting this lot together, the joint prediction over two timesteps is**

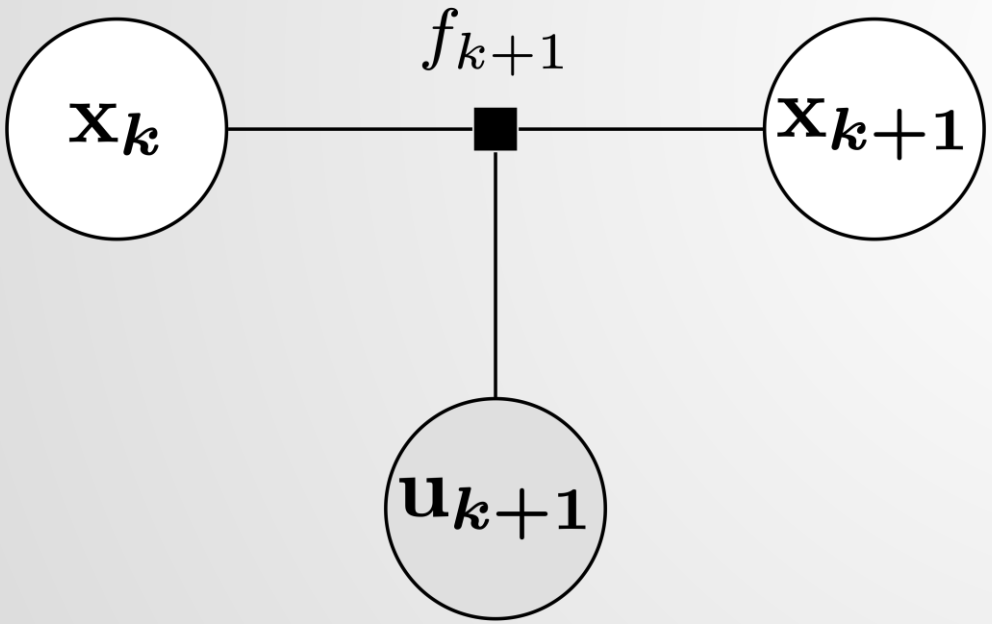
$$f\left(\mathbf{x}_{0:2};\mathbf{u}_{1:2}\right)=f\left(\mathbf{x}_{0}\right)f\left(\mathbf{x}_{1}|\mathbf{x}_{0};\mathbf{u}_{1}\right)f\left(\mathbf{x}_{2}|\mathbf{x}_{1};\mathbf{u}_{2}\right)$$

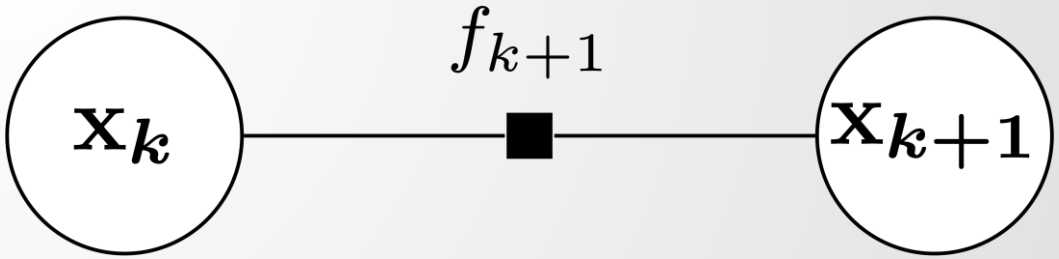
Factor Graph for Time Step 2



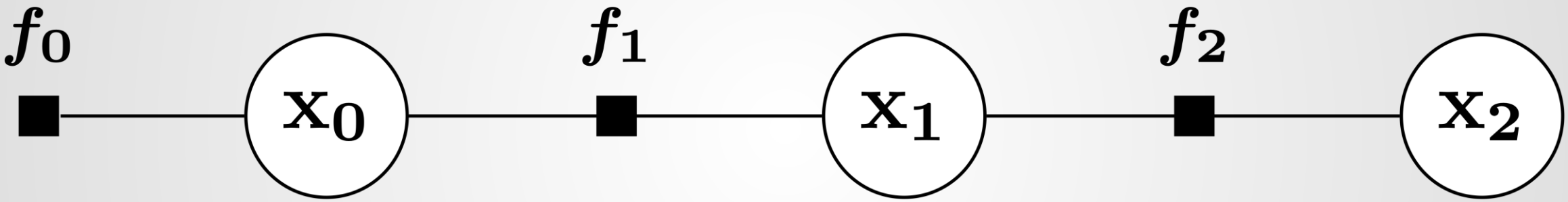
Simplifying Drawing the Factor Graph

- Drawing the control input takes a lot of space
- Therefore, we'll "merge" it into the prediction factor,





Factor Graph for Time Step 2



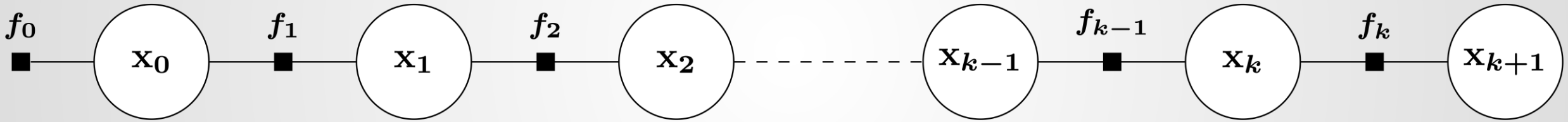
Prediction

- By induction we can expand this to k timesteps to get

$$f\left(\mathbf{x}_{1:k};\mathbf{u}_{1:k}\right)=f\left(\mathbf{x}_0\right)\prod_{i=1}^{k}f\left(\mathbf{x}_i|\mathbf{x}_{i-1};\mathbf{u}_i\right)$$

$$f\left(\mathbf{x}_i|\mathbf{x}_{i-1};\mathbf{u}_i\right)=f\left(\mathbf{v}_i=\mathbf{x}_i-\mathbf{F}_i\mathbf{x}_{i-1}-\mathbf{B}_i\mathbf{u}_i\right)$$

Factor Graph for Predicting $k+1$ Steps



Updating

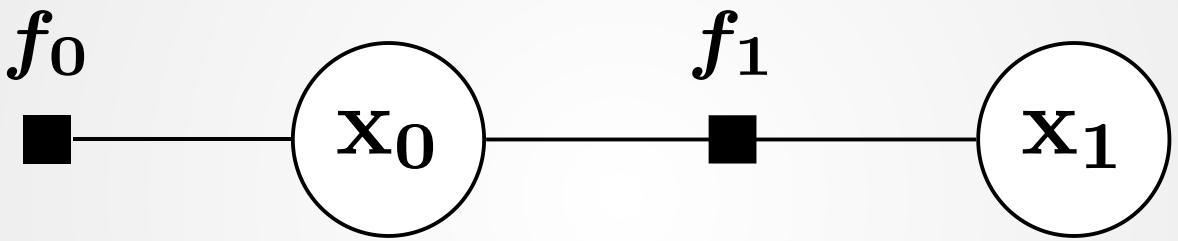
- In the Kalman filter, we combined the predicted estimate with the observation and threw the predicted value away
- Since we want to keep the whole history we don't delete nodes
- Therefore, to do an update all we do is add an extra factor which corresponds to the information fusion process

Fusing Data

- **The observation is fused using Baye's Rule**

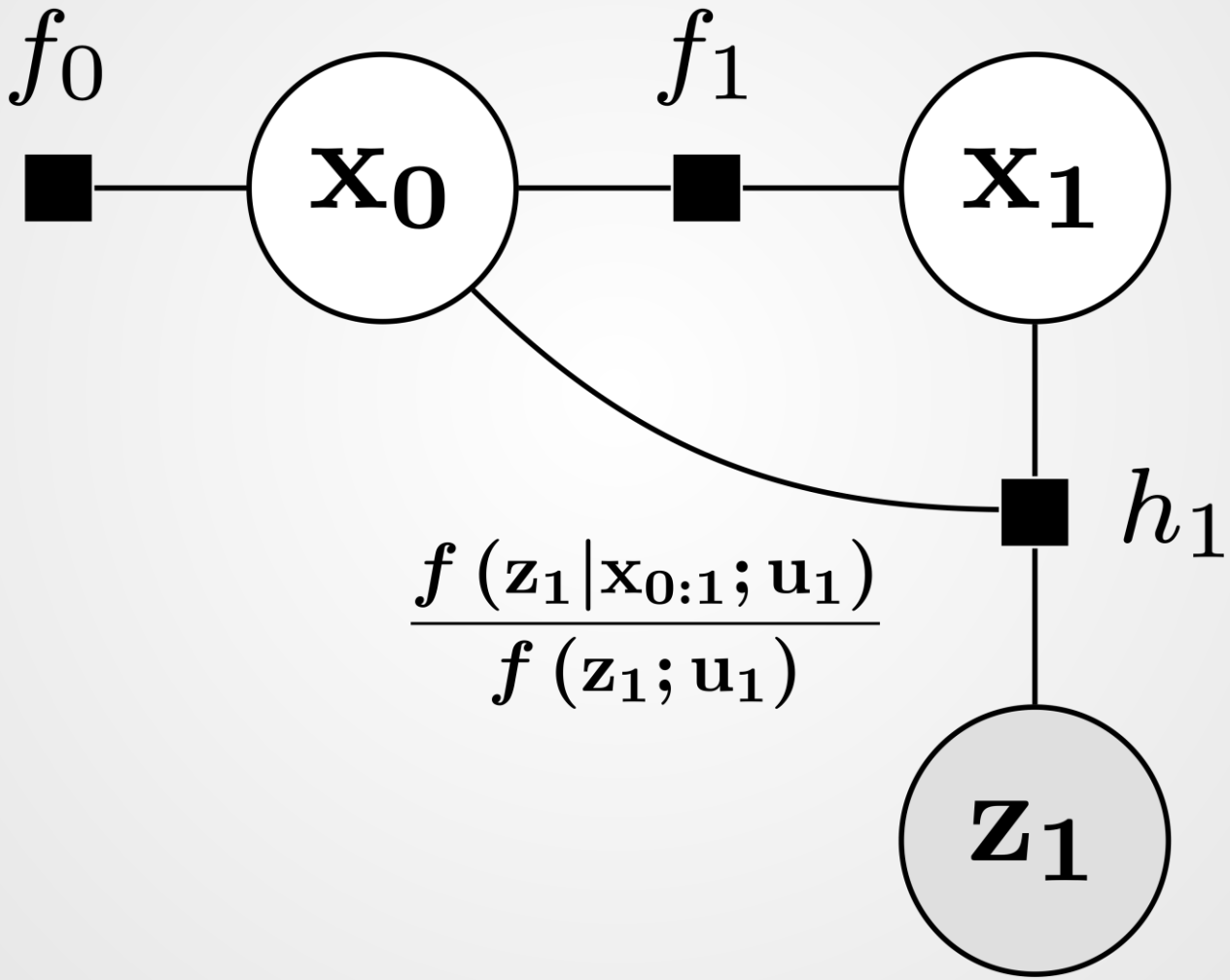
$$f\left(\mathbf{x}_{0:1}|\mathbf{z}_1;\mathbf{u}_1\right)=\frac{f\left(\mathbf{z}_1|\mathbf{x}_{0:1};\mathbf{u}_1\right)}{f\left(\mathbf{z}_1;\mathbf{u}_1\right)}f\left(\mathbf{x}_{0:1};\mathbf{u}_1\right)$$

Graph Before the Update



$$f\left(\mathbf{x}_{0:1}; \mathbf{u}_1\right)$$

Naïve Factor Graph After the Update



Fusing Data

- **The observation model is accounted for through the likelihood term**

$$f\left(\mathbf{z}_1 \middle| \mathbf{x}_{0:1}; \mathbf{u}_1\right)$$

Observation Model and Measurement Likelihood

- Recall the observation model is

$$\mathbf{z}_1 = \mathbf{H}_1 \mathbf{x}_1 + \mathbf{w}_1$$

- Therefore, $\mathbf{z}_1$ *is* a function of just $\mathbf{x}_1$ and $\mathbf{w}_1$
- It *is not* a function of $\mathbf{x}_0$, $\mathbf{u}_1$ or $\mathbf{v}_1$

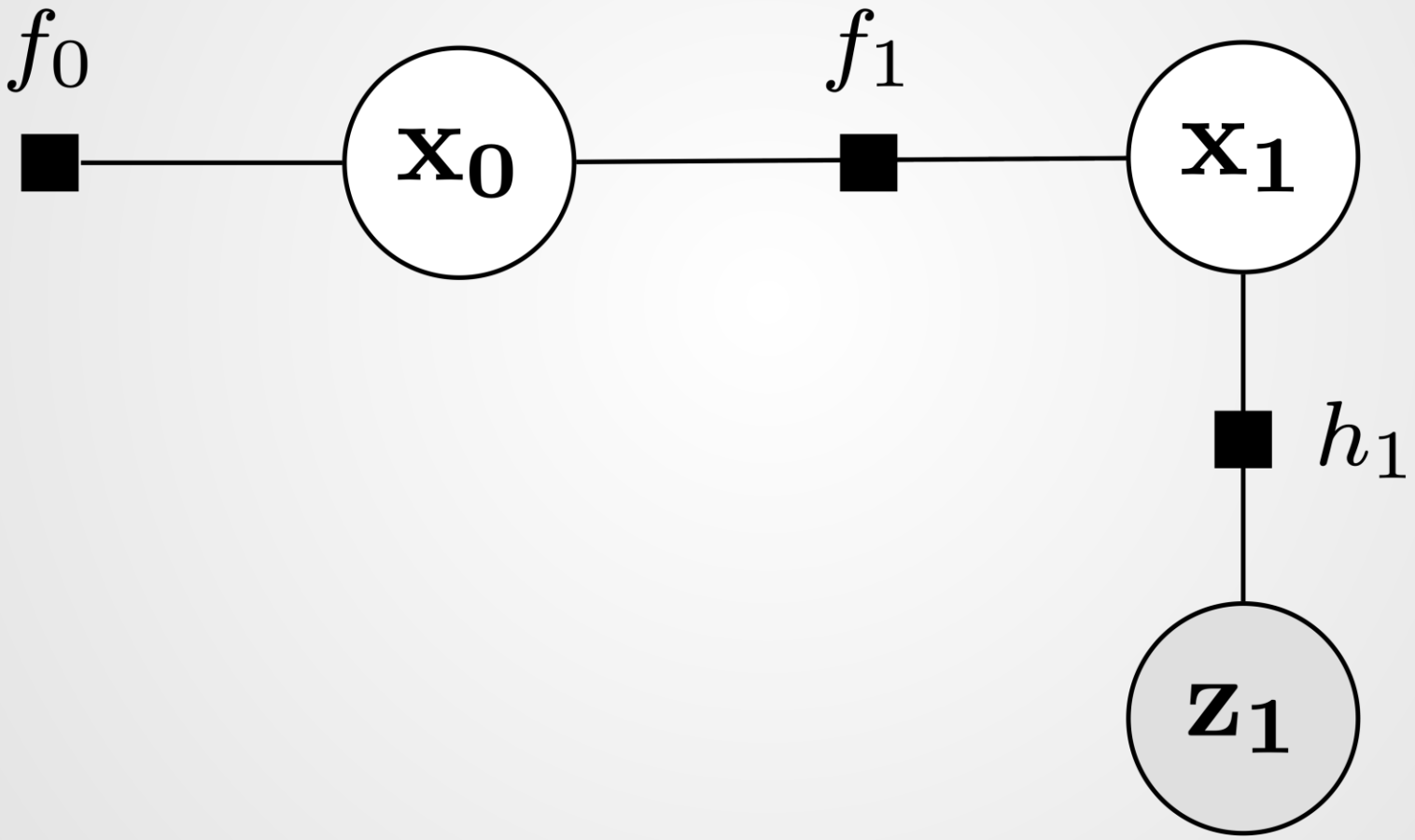
Observation Model and Measurement Likelihood

- **Therefore, we can simplify this to**

$$f\left(\mathbf{z}_1 | \mathbf{x}_{0:1}; \mathbf{u}_1\right) = f\left(\mathbf{z}_1 | \mathbf{x}_1\right)$$

$$f\left(\mathbf{x}_{0:1} | \mathbf{z}_1; \mathbf{u}_1\right) = \frac{f\left(\mathbf{z}_1 | \mathbf{x}_1\right)}{f\left(\mathbf{z}_1; \mathbf{u}_1\right)} f\left(\mathbf{x}_{0:1}; \mathbf{u}_1\right)$$

Factor Graph After the Update



Observation Model and Measurement Likelihood

- Just like the process model, we can rearrange the observation model to make the measurement noise the subject of the formula,

$$\mathbf{w}_1 = \mathbf{z}_1 - \mathbf{H}_1 \mathbf{x}_1$$

Observation Model and Measurement Likelihood

- **The likelihood is computed from**

$$f\left(\mathbf{z}_1 | \mathbf{x}_1\right) = f\left(\mathbf{w}_1 = \mathbf{z}_1 - \mathbf{H}_1 \mathbf{x}_1\right)$$

Probability vs. Likelihood

- We've said that is the observation likelihood

$$f\left(\mathbf{z}_1 | \mathbf{x}_1\right) = f\left(\mathbf{w}_1 = \mathbf{z}_1 - \mathbf{H}_1 \mathbf{x}_1\right)$$

- However, we compute it from the PDF of the observation noise
- What's the difference between the two?

Probability vs. Likelihood

$$f\left(\mathbf{z}_1 \middle| \mathbf{x}_1\right)$$

Observation Model and Measurement Likelihood

- Since $\mathbf{z}_1$ is fixed but $\mathbf{x}_1$ varies, the observation model acts as a *likelihood*
- To remind us of this fact, we will rewrite the likelihood as

$$f\left(\mathbf{z}_1 | \mathbf{x}_1\right) = L\left(\mathbf{x}_1; \mathbf{z}_1\right)$$

Normalization Constant

- The final term we have is the normalization constant
- We originally introduced it as

$$f\left(\mathbf{z}_1;\mathbf{u}_1\right)=\int f\left(\mathbf{z}_1|\mathbf{x}_{0:1};\mathbf{u}_1\right)f\left(\mathbf{x}_{0:1};\mathbf{u}_1\right)\mathrm{d}\mathbf{x}_{0:1}$$

- Substituting what we know about the notation,

$$f\left(\mathbf{z}_1;\mathbf{u}_1\right)=\int\int f\left(\mathbf{z}_1|\mathbf{x}_1\right)f\left(\mathbf{x}_0\right)f\left(\mathbf{x}_1|\mathbf{x}_0;\mathbf{u}_1\right)\mathrm{d}\mathbf{x}_1\mathrm{d}\mathbf{x}_0$$

Observation Model and Measurement Likelihood

- **Substituting all the bits together,**

$$f\left(\mathbf{x}_{0:1}|\mathbf{z}_1;\mathbf{u}_1\right)=\frac{L\left(\mathbf{x}_1;\mathbf{z}_1\right)f\left(\mathbf{x}_0\right)f\left(\mathbf{x}_1|\mathbf{x}_0;\mathbf{u}_1\right)}{\int\int f\left(\mathbf{z}_1|\mathbf{x}_1\right)f\left(\mathbf{x}_0\right)f\left(\mathbf{x}_1|\mathbf{x}_0;\mathbf{u}_1\right)\mathrm{d}\mathbf{x}_1\mathrm{d}\mathbf{x}_0}$$

Normalization Constant

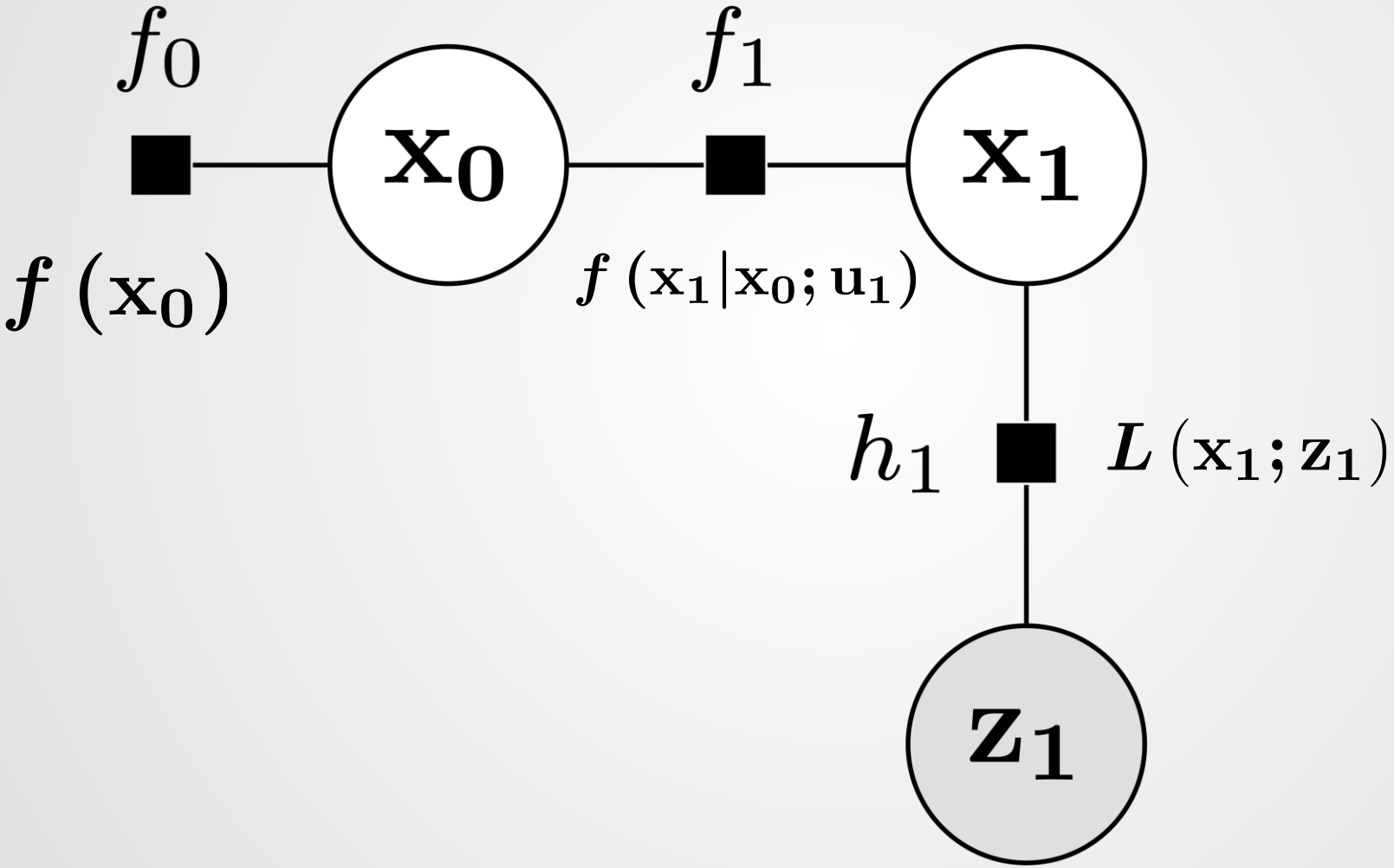
- The normalization constant is also known as the *evidence*
- It tells you how well your model fits the data
- The maximum a posteriori estimator means we can ignore it
- Therefore, we are going to treat it as a constant whose value we don't need to know

Observation Model and Measurement Likelihood

- **Therefore, we can rewrite Baye's Rule as**

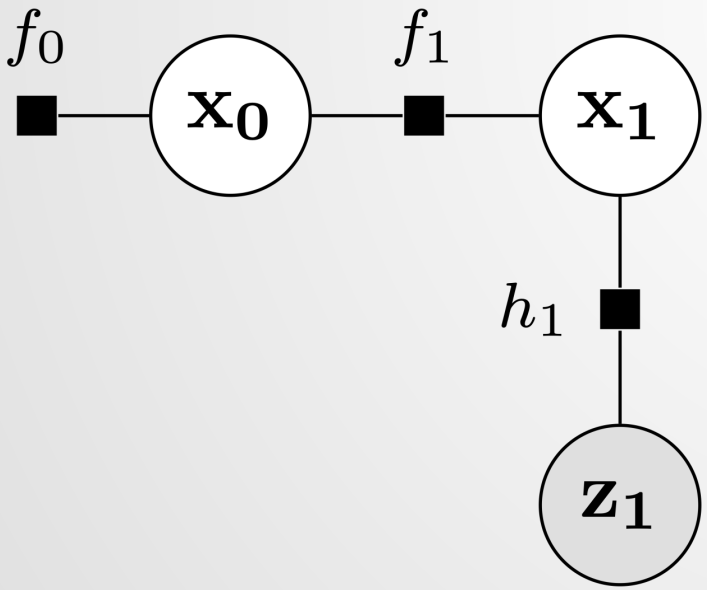
$$f\left(\mathbf{x}_{0:1}|\mathbf{z}_1;\mathbf{u}_1\right) = cL\left(\mathbf{x}_1;\mathbf{z}_1\right) f\left(\mathbf{x}_0\right) f\left(\mathbf{x}_1|\mathbf{x}_0;\mathbf{u}_1\right)$$

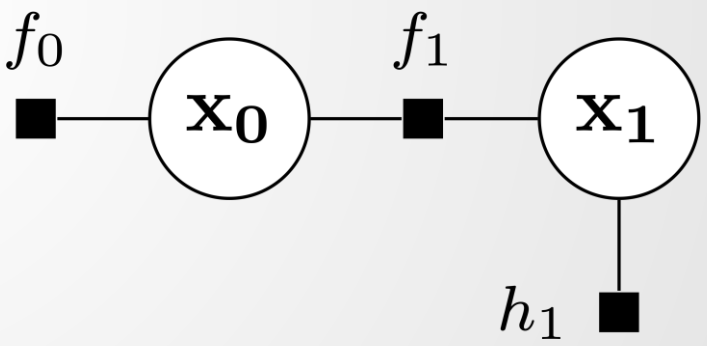
Graph after the Update



More Drawing Simplification

- **Just like the control inputs, we'll simplify the graph for the observation by folding the observations into the observation factor**





Two Timestep Case

- **Now let's consider the case that we want to compute over *three* timesteps with two control input and two observations,**

$$f\left(\mathbf{x}_{0:2} \middle| \mathbf{z}_{1:2}; \mathbf{u}_{1:2}\right)$$

Two Timesteps

- **Using Bayes Rule and ignoring the normalization constant,**

$$f\left(\mathbf{x}_{0:2}|\mathbf{z}_{1:2};\mathbf{u}_{1:2}\right) \propto f\left(\mathbf{z}_{1:2}|\mathbf{x}_{0:2};\mathbf{u}_{1:2}\right) f\left(\mathbf{x}_{0:2};\mathbf{u}_{1:2}\right)$$

Two Timesteps

- **We've already seen that**

$$f\left(\mathbf{x}_{0:2};\mathbf{u}_{1:2}\right)=f\left(\mathbf{x}_{0}\right)f\left(\mathbf{x}_{1}|\mathbf{x}_{0};\mathbf{u}_{1}\right)f\left(\mathbf{x}_{2}|\mathbf{x}_{1};\mathbf{u}_{2}\right)$$

Likelihood Over Two Timesteps

- **We still need to work out the likelihood,**

$$f\left(\mathbf{z}_{1:2} \middle| \mathbf{x}_{0:2}; \mathbf{u}_{1:2}\right)$$

Likelihood Over Two Timesteps

- We can write the observation models down for both timesteps,

$$\mathbf{z}_1 = \mathbf{H}_1\mathbf{x}_1 + \mathbf{w}_1$$
$$\mathbf{z}_2 = \mathbf{H}_2\mathbf{x}_2 + \mathbf{w}_2$$

- $\mathbf{z}_1$ *is* just a function of $\mathbf{x}_1$ and $\mathbf{w}_1$
- $\mathbf{z}_2$ *is* just a function of $\mathbf{x}_2$ and $\mathbf{w}_2$

Likelihood Over Two Timesteps

- **Therefore,**

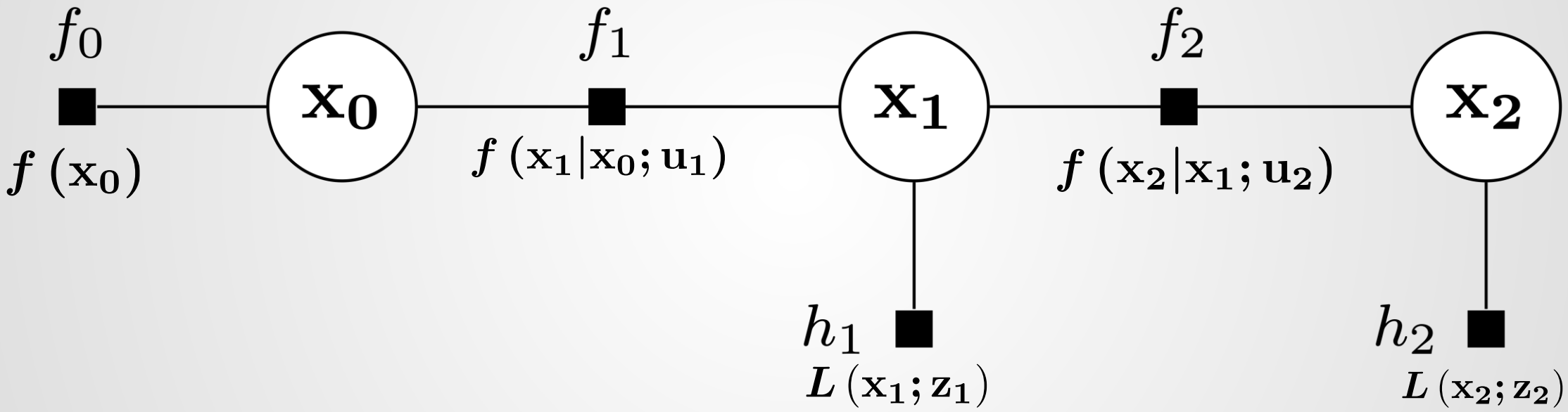
$$
\begin{aligned}
f\left(\mathbf{z}_{1:2} \middle| \mathbf{x}_{0:2}; \mathbf{u}_{1:2}\right) &= f\left(\mathbf{z}_1 \middle| \mathbf{x}_1\right) f\left(\mathbf{z}_2 \middle| \mathbf{x}_2\right) \\
&= L\left(\mathbf{x}_1; \mathbf{z}_1\right) L\left(\mathbf{x}_2; \mathbf{z}_2\right)
\end{aligned}
$$

Finally,

- **Putting everything together, we end up with**

$$
\begin{aligned}
f\left(\mathbf{x}_{0:2} \middle| \mathbf{z}_{1:2}; \mathbf{u}_{1:2}\right) \propto & f\left(\mathbf{z}_{1:2} \middle| \mathbf{x}_{0:2}; \mathbf{u}_{1:2}\right) f\left(\mathbf{x}_{0:2}; \mathbf{u}_{1:2}\right) \\
& = L\left(\mathbf{x}_1; \mathbf{z}_1\right) L\left(\mathbf{x}_2; \mathbf{z}_2\right) \\
& \quad \times f\left(\mathbf{x}_0\right) f\left(\mathbf{x}_1 \middle| \mathbf{x}_0; \mathbf{u}_1\right) f\left(\mathbf{x}_2 \middle| \mathbf{x}_1; \mathbf{u}_2\right)
\end{aligned}
$$

Graph after the Update



Extending to Arbitrary Timesteps

- Not surprisingly, we can follow this pattern
- If we have k timesteps, we have

$$f\left(\mathbf{x}_{0:k}|\mathbf{z}_{1:k};\mathbf{u}_{1:k}\right) \propto f\left(\mathbf{x}_0\right)\prod_{i=1}^{k} f\left(\mathbf{x}_i|\mathbf{x}_{i-1};\mathbf{u}_i\right)$$

$$\times \prod_{i=1}^{k} L\left(\mathbf{x}_i;\mathbf{z}_i\right)$$

Temporal Formulation

- We've seen that we can write a time-based tracking problem in factor graph form
- In the next lecture, we'll see how we can use this to extract state estimates in a fast and efficient manner
- We'll then seen how to apply it to some linear and nonlinear examples